

Estimating Suspended Sediment Concentration using Acoustic Doppler Current Profiler (ADCP)

S. Ferdhous^{1*}, B. Roy² and M. Moniruzzaman¹

Abstract

Suspended sediment concentration (SSC) is an important parameter in the studies of sediment transport and highly important in river management. However, the measurement of SSC still uses conventional techniques. These techniques have limitations, especially in temporal resolution. With advanced technology, the measurement of sediment concentration can be done using hydroacoustic technology such as the Acoustic Doppler Current Profiler (ADCP). The signal-to-noise ratio (SNR) data of ADCP can be used to indicate sediment concentration in the river water. This study evaluated SNR data from ADCP to estimate SSC in the branch channel to the river Padma. The frequencies of ADCP used in this study were 1 MHz and 3 MHz. Calibration and validation of SSC_{ADCP} have been achieved using a linear regression model, considering SSC_{meas} as an independent variable. The linear regression model for the validation of SSC_{ADCP} showed a good relationship with $R^2 = 0.97$ for cross-section 01 and $R^2 = 0.94$ for cross-section 02, respectively. This technique can be highly useful for practical applications, particularly in scenarios where collecting sediment samples is challenging. By utilizing ADCP-based hydroacoustic technology, real-time and continuous monitoring of suspended sediment concentration becomes feasible, overcoming the limitations of conventional sampling methods.

Keywords: ADCP, Suspended sediment concentration, Signal to Noise Ratio (SNR), Hydroacoustic sediment monitoring, Sediment transport measurement.

Introduction

Fluvial river systems are used for numerous purposes like water supply, hydroelectric power generation, transportation etc. all over the world. The production of the sediment is one of the effects caused by the use of these systems in the basin (Wosiacki et al., 2018). Sediment is an integral part of river system and plays very important role in the context of sustainability of riverine environments (Bečvář, 2006). Suspended sediment concentration (SSC) is an important parameter in the studies of sediment transport and highly valuable in overall river management and planning (Dwinovantyo et al., 2017; Woonchul et al., 2022). The monitoring of fluvial suspended sediment transport is much needed for the assessment of hydro-morphological processes, river environments and many social activities associated with river management (Flóra and Baranya, 2020).

The measurement of SSC in the water column can be conducted in several ways, such as conventional, optical, and acoustic methods (Dwinovantyo et al., 2017). The conventional gravimetric method for the assessment of SSC in laboratory is relatively difficult to achieve because it requires collection of numerous water samples in each point of interest with simultaneous lack of continuous temporal and spatial data (Zhang et al., 2013). Slow and relatively higher measurement cost is another disadvantage of conventional gravimetric method (Holdaway et al., 1999). The measurement of flow velocities in order to determine river discharge is the main application field of Acoustic Doppler current profilers (ADCPs) (Baranya and Józsa, 2013). Over the past two decades, the development of hydroacoustic instrument such as

the Acoustic Doppler Current Profilers (ADCP) for measuring water velocity by detecting acoustic energy backscattered from suspended matter in the water column has permitted the detection of suspended sediment with very high temporal and spatial resolution. It is also a substantial advantage for sediment flux estimation that these instruments can return data of both velocity and sediment concentration (Venditti et al., 2016). Recent studies demonstrated good correlation between acoustic backscatter from ADCP and measured concentration by direct measurement (Guerrero et al., 2016). Many researchers also used numerical analysis to compare the measured SSC from ADCP and sediment transport model/numerical model (Bayram et al., 2012). This paper describes the results of a field experiment where SSC were estimated at multiple layers of depth from the ADCP. The aim of this study was to compare laboratory measured SSC data to SSC data derived from SNR data of ADCP.

Methods

Theoretical Background

The method of conversion of Signal to Noise Ratio (SNR) data into SSC can be done based on the principle of sonar equation (Wall et al., 2006). The direct measurement of SSC can be correlated with SSC data derived from SNR data of ADCP using simple regression linear model. After that, SNR value from the ADCP can be converted into SSC in $mg\ L^{-1}$. The relationship between SSC and the relative SNR can be expressed as (Gartner and Ganju, 2007)

¹Geotechnical Research Directorate, River Research Institute (RRI), Faridpur-7800, Bangladesh.

*Corresponding Author: (E-mail: sumiya_eee@yahoo.com)

²Hydraulic Research Directorate, River Research Institute (RRI), Faridpur-7800, Bangladesh.

$$SSC = 10^{(A+B \cdot SNR)}$$

Eq. (1)

where A and B in Eq. (1) are empirical parameters that can be derived from known SSC and SNR data pairs using e.g., least squares fitting.

The corrected signal is the measurement of SNR corrected for transmission losses in units of dB. The corrected signal is computed using the following equation.
Corrected Signal = $SNR_{mean} + 10\psi \log_{10}(R) + R_{aw} +$

R α_s

Eq. (2)

where SNR_{mean} is the mean of SNR of the 4 beams of the ADCP along of each vertical, $10\log_{10}(\psi R)$ is the spread attenuation, R_{aw} is the water absorption R_{as} is the sediment attenuation term. Also, R is the slant range from transducer head to measured beam (m). When calculating the effect of spherical spreading close to the transducer, a near-field correction factor ψ has to be introduced, according to Downing et al. (1995). This correction can be calculated based on the critical range $R_{critical}$, where $R_{critical} = \pi a_t^2 / \lambda$. Here a_t is the transducer radius in cm, and λ is the acoustic wavelength. The correction factor for near-field spreading loss is

$$\psi = \frac{1 + 1.35Z + (2.5Z)^{3/2}}{1.35Z + (2.5Z)^{3/2}}$$

Eq. (3)

where Z equals to $R/R_{critical}$.

In this study, the above presented correction factor was used for near-field calculations of spreading losses.

The speed of sound had been computed by the following equation

$$c = 1404.3 + 4.7T - 0.047T^2$$

Eq. (4)

where T is the water temperature, in °C.

For the estimation of energy dissipation by absorption in the water, we used the formula of Schulkin and Marsh (1962) as follows:

$$\alpha_w = 8.687 \frac{3.3810^{-6} f^2}{6 - \left[\frac{1520}{21.9 \times 10^{273+T}} \right]}$$

Eq. (5)

where f is the acoustic frequency, in kilocycles per second, T is the water temperature, in °C.

Attenuation from suspended sediment (α_s) is caused by both scattering and absorbing the energy. It is shown that energy dissipation depends mainly on particle size and sound frequency (DRL Software Ltd, 2003), and under certain conditions one or both of them can be neglected. We did not consider attenuation due to absorption by the sediment here.

Study Site

The study site is located at North-East of Faridpur Town. A branch channel of the Padma River is flowing there. The study site is situated in the 45Q UTM zone. It lies between 2615000 mN to 2625500 mN and 788700 mE to 794900 mE (**Fig. 1**).

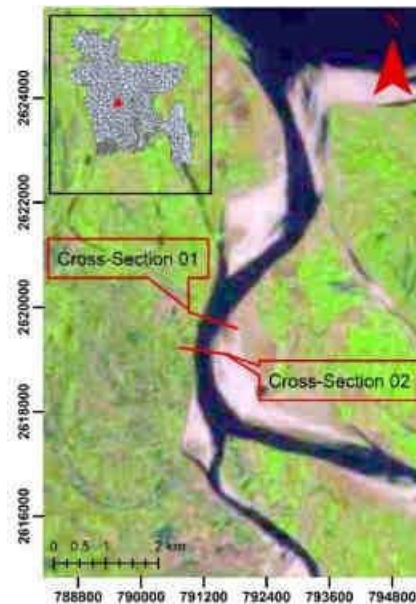


Fig. 1. Location map of Study Site

Materials and equipment

two cross-sections with an average distance of 600

Equipment used in this study was Sontek M9 ADCP. The survey, which involved moving boats operating in

meters between them, was conducted in January 2023. All water samples were preserved in pre-cleaned

polyethylene bottles. Water samples were collected in 1-liter bottles from 10 different depths, starting from 2 meters until 10 meters for each 2-meter interval. The amount of water sample was 1 liter. To avoid loss of water volume due to evaporation, the bottles were

stored in a cooler box with ice cube ($< 4^{\circ}\text{C}$) before analyzing in the laboratory.

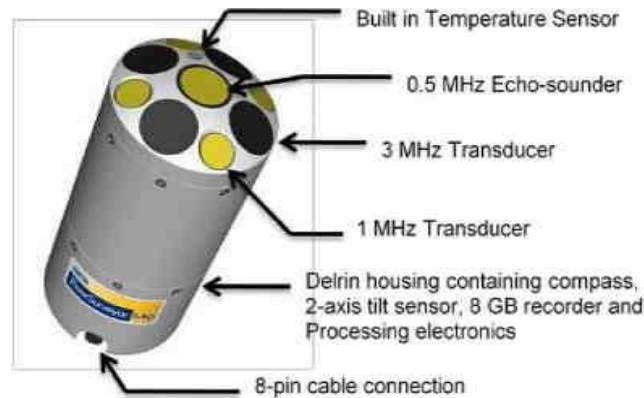


Fig. 2. Different features of Sontek M9 ADCP

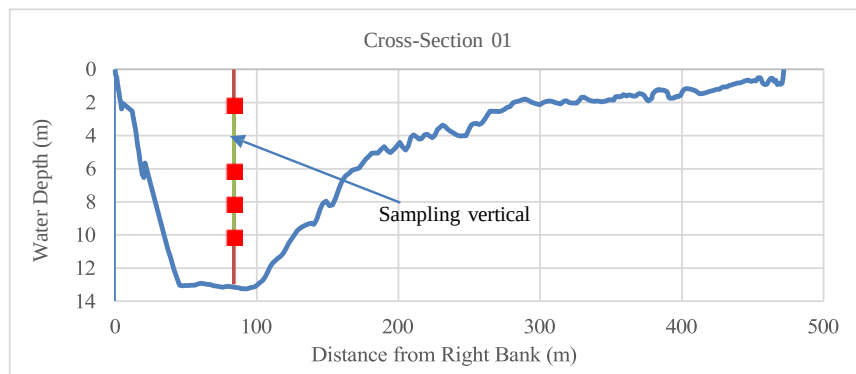


Fig. 3. Cross sectional profile (Section-1)

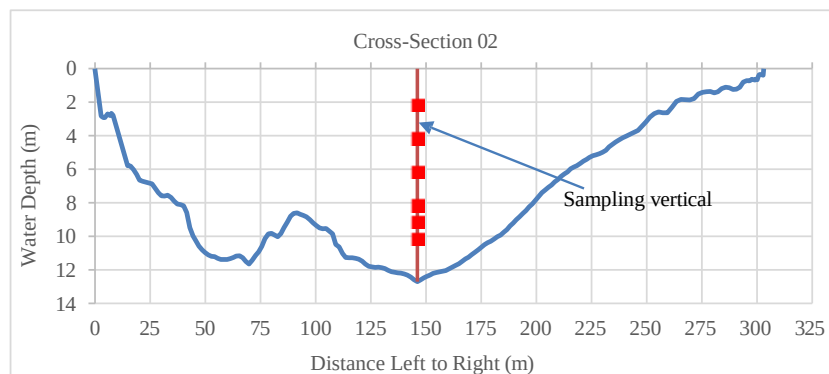


Fig. 4. Cross sectional profile (section-2)

Laboratory Analysis

Suspended sediment concentration of ten (10) river water samples were analyzed in the Sediment Laboratory at River Research Institute, Faridpur. Suspended sediment portion of each sample was separated from the liquid portion by using filter paper (Whatman, 42 ASHLESS). Filter papers with the retained materials were oven dried. The 1 liter of water was then filtered using filter paper (Whatman, 42 ASHLESS). Before the filtration process, the filter

paper was dried at a temperature of 105°C for 1 hour and then the filter paper was cooled in a desiccator and weighed. After water filtration process, filter paper plus residue was dried again for two hours at 105°C (Flóra Pomázi, 2020).

Estimation of suspended sediments concentration

For ADCP-based SSC estimate, direct measurement of SSC with known concentration values requires

calibration. The following formula used to calculate the direct SSC measurement (mg L^{-1})

$$SSC = \frac{(A-B) \times 1000}{\text{Sample Volume, mL}} \quad \text{Eq. (6)}$$

where, A = final weight of filter + dried residue in mg, B = weight of filter in mg.

For final conversion of SSC from $SNR_{corrected}$ value the following relationship $SSC = 10^{(A+B \cdot SNR_{corrected})}$ was used. The values of empirical parameters A and B were determined using least squares fitting between data pairs of SSC_{meas} and $SNR_{corrected}$. And for our case

$SNR_{corrected}$ was computed using the following equation-

$$SNR_{corrected} = SNR_{mean} + 10\psi \log_{10}(R) + R\alpha_w + R\alpha_s \quad \text{Eq. (7)}$$

Results and Discussion

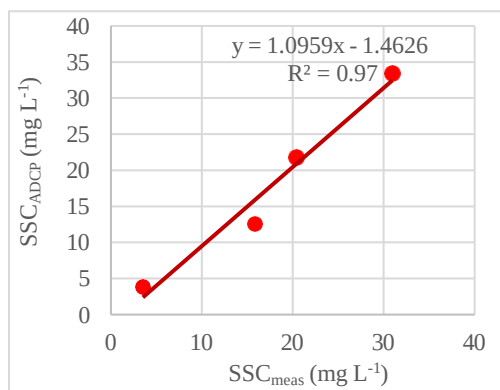
Calibration curve was built to relate $SNR_{corrected}$ values to SSC_{meas} . The direct measurement of suspended sediment concentration were matched with $SNR_{corrected}$ values from the corresponding depth layer or bin to establish the calibration relationship (Dwinovantyo, 2017). For creating calibration curve, SSC_{meas} was plotted as ordinate and $SNR_{corrected}$ was plotted as abscissa (Venditti et al., 2016).

Table 1. Sample collecting depth with corresponding SNR, SSC_{meas} , SSC_{ADCP} values for Cross-Section 1

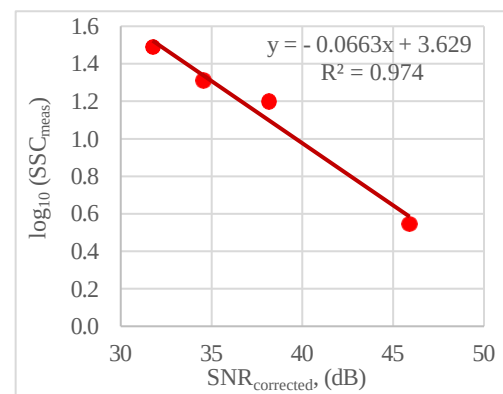
Depth (m)	$SNR_{corrected}$ (dB)	SSC_{meas} (mg L^{-1})	$\log_{10}(SSC_{meas})$	SSC_{ADCP} (mg L^{-1})
2.14	45.90	3.51	0.55	3.85
6.14	38.14	15.81	1.20	12.60
8.14	34.56	20.41	1.31	21.77
10.14	31.76	30.95	1.49	33.38

Fig. 5 (a) displays the results least squares fitting between data pairs of $SNR_{corrected}$ and measured suspended sediment concentration (SSC_{meas}) $\log_{10}(SSC_{meas})$. It can be seen that empirical parameter values A = 3.629 and B = - 0.0663 with an acceptable fit of coefficient of determination $R^2 = 0.974$ for cross-

section 1. According to **Fig. 5 (b)** the relation between SSC_{ADCP} and SSC_{meas} has strong correlation which R-squared value is 0.97. In such a case Dwinovantyo, et al., (2017) found R-squared value = 0.7196 between SSC_{ADCP} and SSC_{meas} at Lembbeh Strait, North Sulawesi, Indonesia.



(a)



(b)

Fig. 5. (a) Calibration and (b) Validation of SNR using direct sample measurement for the samples collected from cross-section 1

Table 2. Sample collecting depth with corresponding SNR, SSC_{meas} , SSC_{ADCP} values for Cross-Section 2

Depth (m)	$SNR_{corrected}$	SSC_{meas} (mg L^{-1})	$\log_{10}(SSC_{meas})$	SSC_{ADCP} (mg L^{-1})
2.14	55.80	4.1	0.61	3.84
4.14	41.22	12.6	1.10	15.75
6.14	40.21	18.2	1.26	17.35
8.14	37.26	21.0	1.32	23.10
9.14	35.98	28.2	1.45	26.13
10.14	35.08	32.8	1.52	28.51

Similarly for cross-section 2 least squares fitting had been performed between data pairs of $SNR_{corrected}$ and measured suspended sediment concentration (SSC_{meas}) (**Fig. 6**). This analysis yields the empirical parameter values $A = 2.9284$ and $B = -0.042$ with an acceptable

fit of coefficient of determination $R^2 = 0.968$ for cross-section 2. Results of the acoustic method compared to direct measurement shows good agreement showed at **Fig. 6** by $R^2 = 0.9439$ for cross-section 2.

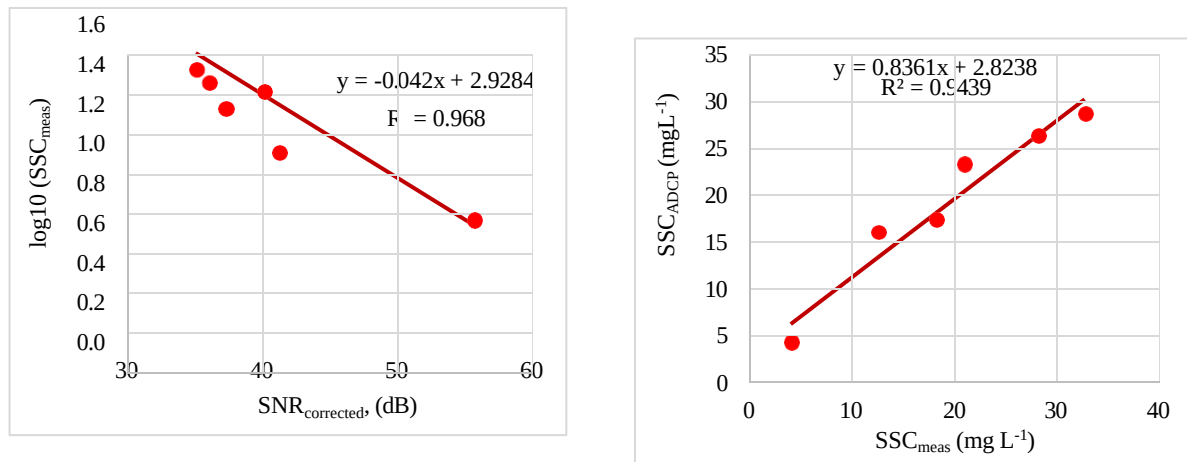


Fig. 6. (a) Calibration and (b) Validation of SNR using direct sample measurement for the samples collected from cross-section 2

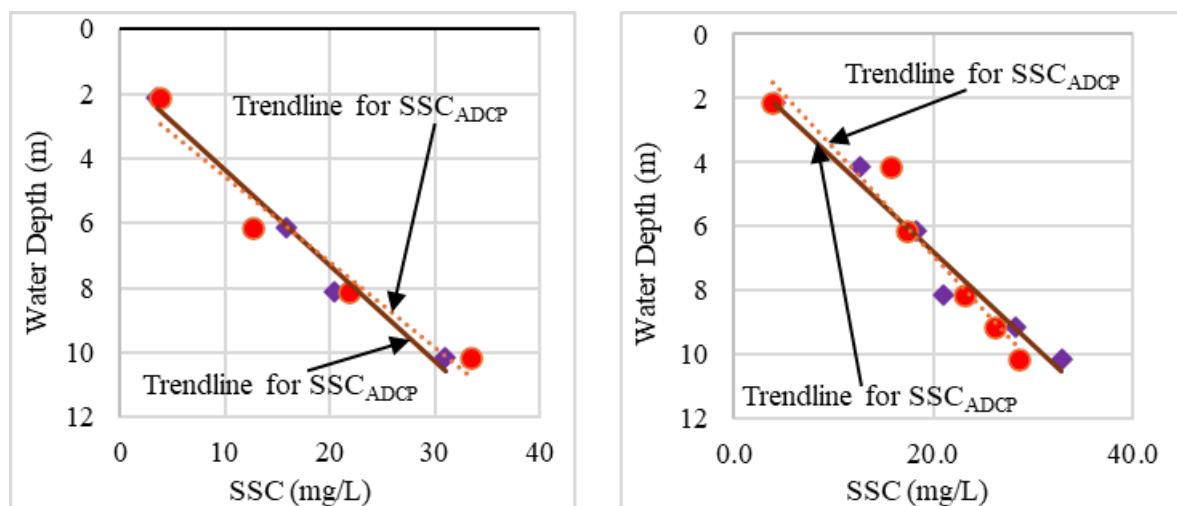


Fig. 7. Depth vs SSC relationship

Another key observation is the depth-dependent variation in SSC. As shown in **Tables 1** and **2** and illustrated in **Fig. 7**, SSC generally increases with depth. This pattern is consistent with natural fluvial dynamics, where coarser or denser particles tend to settle at deeper layers due to gravity and reduced turbulence. The ADCP-based estimates captured these vertical trends effectively, affirming the device's capability to resolve depth-stratified sediment profiles with high resolution.

The implications of these results extend beyond mere accuracy. Utilizing ADCP for sediment concentration monitoring offers a scalable and efficient alternative to conventional sampling, which is labor-intensive and often spatially limited. The ability to derive SSC from

SNR in near real-time could be invaluable for sediment budgeting, flood prediction, infrastructure design, and habitat assessment in dynamic river systems like the Padma.

Moreover, despite the study's modest scale—limited by vessel type and sample size—the high correlation values suggest that the method is robust and adaptable to varying field conditions. This strengthens the case for broader adoption of ADCP technology in sediment monitoring, especially in regions with limited access to advanced sampling infrastructure.

Overall, the results demonstrate that SNR data from ADCP not only correlate strongly with direct SSC measurements but also reflect physically meaningful sediment transport processes. These findings support

the feasibility of integrating ADCP-based SSC estimation into routine river monitoring protocols, enhancing both efficiency and data richness.

Limitations

The limitation of the current study is that the sample size is not large. In this study, the usual motorized boat was used as a survey vessel, not a specialized one suitable for survey purposes. It is very difficult to maintain a particular transect (cross-section line) for the ADCP survey and sample collection using the usual motorized boat.

Conclusion

Suspended sediment concentrations could be estimated using the Signal-to-Noise Ratio (SNR) data. The SNR results confirmed a comparable estimation performance with respect to those given by direct measurement of SSC. The error of estimation between ADCP and direct measurement was not more than 5 mgL⁻¹. The monitoring of SSC with ADCP could be considered a good method to observe time series data of SSC, where it is usually not possible with conventional measurement. Although ADCP shows potential for estimating SSC quickly, several issues remain to be explored, such as the Rayleigh parameter for different grain sizes of suspended sediment, a better understanding of instrument error, and the need for a water sample analysis to calibrate the sonar equation. However, we are sanguine that this study will usher in new prospects for the estimation of sediment concentration using SNR data from ADCP.

Conflict of Interest

The authors declare no conflict of interest.

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